Rafi A. Morshed

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Education

Heritage School
Intl. Advanced Level (IAL), Edexcel, Pearson, UK — Mathematics, Physics, Biology, Chemistry (Expected Oct. 2026)
International General Certificate of Secondary Education (IGCSE), Edexcel, Pearson, UK *Nov. 2024*

Research & Publications

Student Researcher & Co-Author—Human Genetics Research and Training Center (HGRTC)
Dec. 2024 – May 2025 (6Mos)

- Organized clinical sample data, performed statistical data analysis, and created data visualizations in Python for a five-year human papillomavirus epidemiology study (code available on GitHub).

Publication: M. Arifur Rahman, M. Rafiqul Islam, M. Nabil Hasan, M. Sefaut Ullah, Israt Jahan, **Rafi A. Morshed**, et al. (2025). Epidemiology, genotypic diversity and age-specific risk pattern of circulating human papillomavirus in Dhaka City, Bangladesh: a five-year study (2020–2024). *International Journal of Research in Medical Sciences*, 13(7), 2781–2787. DOI ↗

Conference Presentations

Morshed, R., Islam, M., Rahman, M., & Aslam, M. (2026). Epidemiology, genotypic diversity, and age-specific risk pattern of circulating human papillomavirus in Dhaka City, Bangladesh: a five-year study (2020–2024). **EUROGIN 2026, International Multidisciplinary HPV Congress**, Vienna, Austria. [Abstract #12369 ↗]

Independent Projects

Stokes' Law and the Erythrocyte Sedimentation Rate (ESR) Aug. 2026 – Present

- Derived and numerically modeled RBC sedimentation using Stokes' Law (ODEs, Python/SciPy), experimentally verified viscosity via falling-ball method, then applied the model to explain ESR as a clinical inflammation marker — integrating Physics, Maths, Biology, and Chemistry; built the simulation in Python (GitHub).

Leadership, Teaching & Volunteering

Heritage School, Narayanganj, Bangladesh — unless otherwise noted
Founder & Student Lead — Heritage Pathshala Sustainability Initiative Oct. 2025 – Jul. 2026 (10 mos.)

- Founded and led a 32-student volunteer team to design and launch a sustainable funding model for Heritage Pathshala, a free primary school supporting 38 underprivileged children at risk of child labor, drawing on prior leadership experience from the school's Food Bank project.
- Organized 3 alumni outreach seminars on program impact and financial transparency, raising BDT 672,000 (around 5,500 USD) and growing the alumni donor base from 71 to 82 contributors between Jan. and Jun. 2026.
- Introduced a twice-daily meal program that supported student enrollment growth from 38 to 54 and improved retention; built a self-sustaining framework through alternate-year alumni giving, annual donor recognition, and integration with school administration.

Exam-Prep Resource Development (Volunteer) **Portal ↗** *Independent* Feb. – Jun. 2026 (5 mos.)

- Developed and freely distributed exam-prep resources — subject-specific study guides and rapid revision summaries — for IGCSE and IAL students, based on independent analysis of recurring question patterns and past-paper trends.

Teaching Assistant — Physics & Mathematics, Grades 9–10 Jan. 2025 – Dec. 2025 (12 mos.)

- Assisted the teacher in Grade 9 and Grade 10 physics and mathematics classes (26 and 21 students respectively).
- Held weekly 90-minute help sessions for each class, guiding students through challenging problems step by step.
- Graded practice papers, mock questions, and homework sets under teacher supervision.

Professional Development

- Illumina — Introduction to Next-Generation Sequencing: DNA, 10.5 hrs, Jan. 2026 — NGS workflow, sequencing principles, DNA library preparation, whole-genome sequencing applications. Credential ↗
- Hands-on Training on Sanger Sequencing, HGRTC, Dec. 2024 — DNA extraction, amplification, and sequencing analysis.
- Hands-on Training on Real-Time PCR, HGRTC, Mar. 2025 — qPCR principles and data interpretation.

Awards & Honors

- Edexcel International High Achievers' Award, 2025.
- The Daily Star Award, 2025.
- Participant, XXXIII International Philosophy Olympiad, Bari, Italy, 2025. Official Record ↗